Assignment 1: Characterising a SAMBA AMOC Time Series

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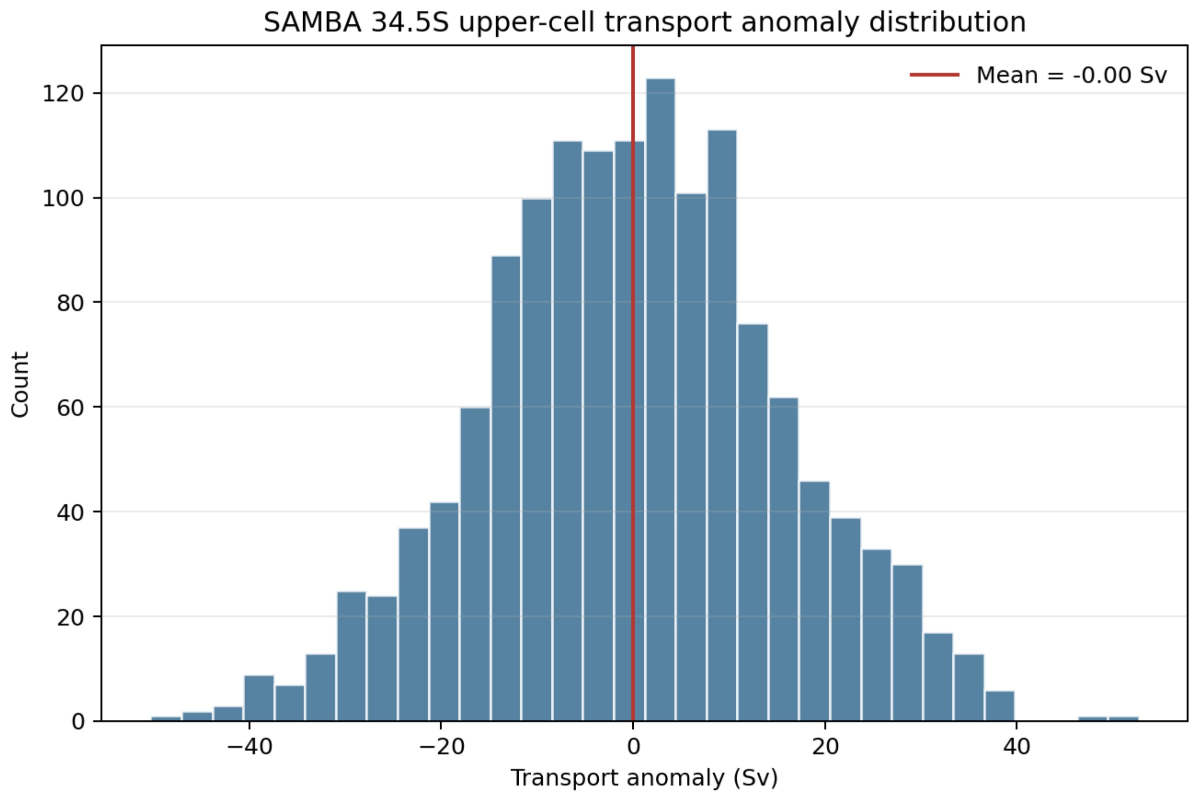
Data choice

This analysis uses the SAMBA 34.5S observing array from AMOCatlas and the UPPER_TRANSPORT variable. This series is the upper-cell volume transport anomaly in Sverdrups (Sv). I chose SAMBA because it provides an AMOC-related time series from the South Atlantic and does not duplicate the MOCHA/RAPID 26N dataset assigned to another student.

The record spans 2013-09-12 to 2017-07-16. It contains 1,404 daily samples, with a median sampling interval of 1.0 day. There are no missing values in the standardised series, so no interpolation was needed. The gap-filling step in the script is kept for reproducibility, but it leaves this particular series unchanged.

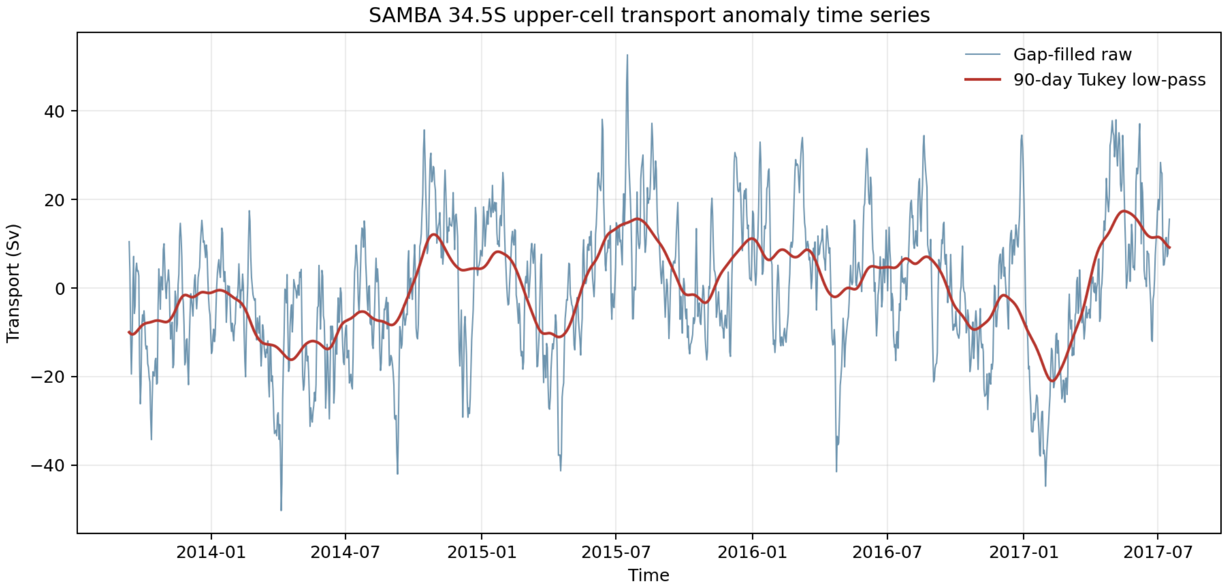
Time-domain statistics

The upper-cell transport anomaly has a mean of -0.003 Sv, which is essentially zero as expected for an anomaly series. The standard deviation is 15.51 Sv, showing substantial variability around the mean. The median is 0.35 Sv. The minimum and maximum are -50.28 Sv and 52.69 Sv, giving a total range of 102.97 Sv. The histogram shows that most values are clustered near zero, but the series also includes large positive and negative transport anomalies.



SAMBA upper-cell transport distribution

The raw time series contains strong day-to-day and multi-week variability. I applied a 90-day Tukey-window low-pass filter using a centred 91-sample rolling window. The filtered curve removes much of the short-timescale variability and highlights slower changes in the upper-cell transport anomaly.



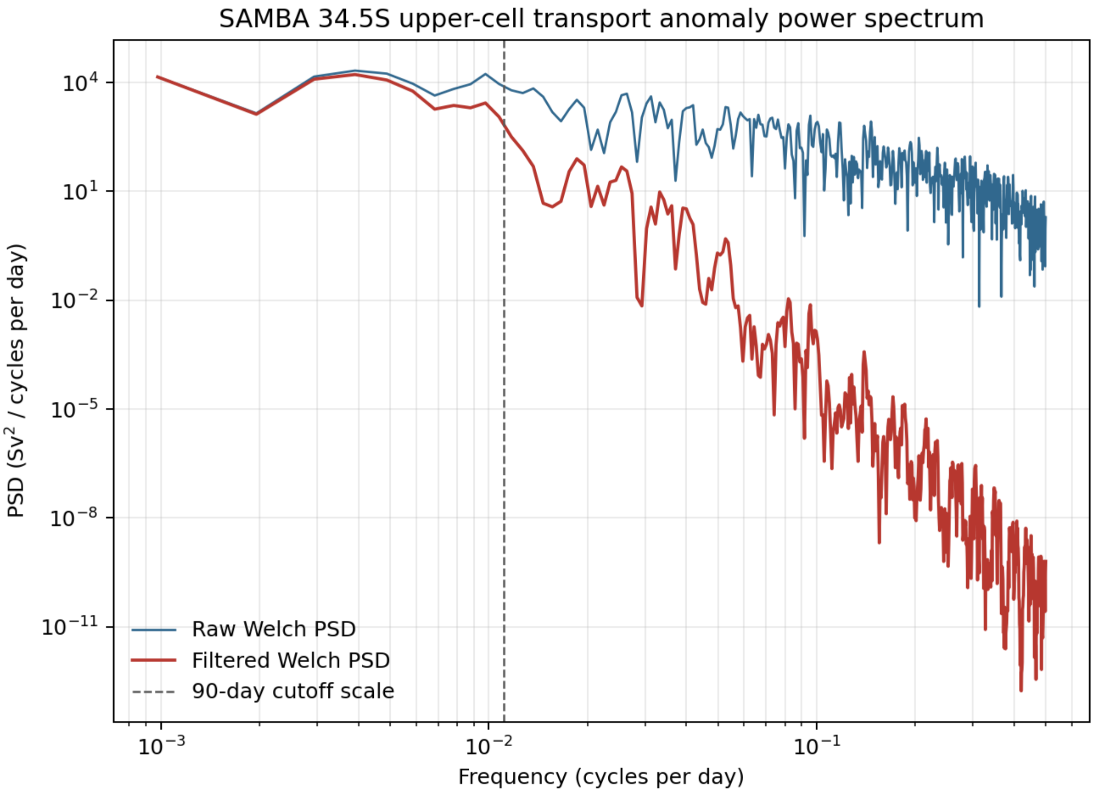
Raw and filtered SAMBA upper-cell transport

Frequency-domain statistics

I estimated the power spectral density using Welch's method with Hann windows, 1024-sample segments, and 50 percent overlap. Since the data are daily, each segment spans 1024 days. This segment length is long enough to retain useful low-frequency resolution for the relatively short SAMBA record, while still using overlapped segment averaging rather than a single raw periodogram.

The integrated Welch spectrum satisfies the Parseval variance check. The ratio between integrated PSD and time-domain variance is 1.002 for the raw series, which is very close to 1. This indicates that the spectral normalisation is consistent with the variance of the time series. The largest spectral peak in this estimate corresponds to a timescale of about 256 days. Overall, the spectrum has stronger low-frequency than high-frequency variance, suggesting a red spectrum with variability concentrated at multi-month and longer timescales.

The 90-day Tukey low-pass filter reduces spectral power at frequencies above about 1/90 cycles per day. The filtered spectrum therefore lies below the raw spectrum at higher frequencies, while retaining more of the lower-frequency variance. This behaviour is consistent with the goal of isolating slower AMOC transport variability.



Raw and filtered SAMBA upper-cell spectrum

Reproducibility

The analysis can be reproduced with:

```
python assignment_analysis.py
pytest -q
```

The script downloads/loads the SAMBA dataset through AMOCatlas, writes the figures to figures/, and writes the numerical summary to outputs/assignment1_summary.json.